

QAF Technical Supplement E:

The Emergence of Quantum Mechanics and General Relativity from the $Sp(2)$ Tensor

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Abstract

We derive the two pillars of modern physics not as independent theories, but as distinct asymptotic limits of a single **Stationary Topological Solid** defined by the $Sp(2)$ automorphism. We demonstrate that Quantum Mechanics (QM) emerges as the **Linear Latency Limit** of the lattice update cycles, identifying "time" as the processing impedance of non-commutative torsion. Conversely, General Relativity (GR) emerges as the **Elastic Strain Limit** of the tensor network, where the Einstein Field Equations represent the generalized **Hooke's Law** for the vacuum stiffness. Crucially, we reject the fluid-dynamic paradigms of expansion and singularities; instead, we define the Cosmos as a global adiabatic system where **Saturated Topological Condensates** (formerly "Black Holes") recycle information via a **Gravitational Meissner Effect**, maintaining an eternal, constant-entropy state.

1 Introduction: The Monist Premise

Standard physics treats Spacetime (GR) and the Quantum State (QM) as separate entities with incompatible mathematical structures. QAF posits a single object: the Autocontained Tensor $\mathcal{Q} \in Sp(2)$.

- **Microscopically (Local):** The algebra is non-commutative. The observer perceives this torsion as Quantum Uncertainty.
- **Macroscopically (Global):** The network has entropy. The observer perceives the gradient of this entropy as Gravity.

2 Derivation I: Quantum Mechanics (The Lattice Latency)

Standard Quantum Mechanics assumes time is a fundamental, continuous parameter flowing externally. In QAF, time is derived as the **Processing Latency** required to resolve algebraic conflicts in the non-commutative lattice. We derive the Schrödinger Equation not as movement in time, but as the sequential updating of the tensor state to maintain unitary consistency.

2.1 1. The Origin of the Wavefunction (Projection)

The fundamental state is a quaternion $q \in \mathbb{H}$. However, a physical observer is defined by a scalar causality limit c (Real axis). The observer cannot access the full tensor state simultaneously. The

"Wavefunction" ψ is the projection of the global tensor state $\mathcal{Q}$ onto the complex sub-manifold accessible to the observer's bandwidth:

$$\psi(\tau) \equiv \text{Proj}_{\mathbb{C}}(\mathcal{Q}) \quad (1)$$

Where τ is the internal update count (the "tick") of the lattice.

2.2 2. Uncertainty as Algebraic Torsion

Heisenberg's Uncertainty is usually interpreted as a limit of knowledge. In QAF, it is the geometric necessity of the non-commutative basis:

$$[\mathbf{i}, \mathbf{j}] = 2\mathbf{k} \neq 0 \quad (2)$$

This commutator represents the **Algebraic Torsion**. To measure property A (axis $\mathbf{i}$) and then property B (axis $\mathbf{j}$), the lattice must perform a rotation operation. This operation is not instantaneous; it has an "impedance." The Planck constant $\hbar$ is the quantum of this rotational action.

2.3 3. Time as Impedance (Deriving Schrödinger)

Since the basis vectors do not commute, the lattice cannot update all links simultaneously without violating logic. It must serialize the updates. The evolution operator $U(\delta\tau)$ is the rotation required to resolve the torsion generated by the Hamiltonian density $\hat{H}$ (the stress energy of the defect).

$$\mathcal{Q}(\tau + \delta\tau) = \left(\mathbb{I} - \frac{\mathbf{i}}{\hbar} \hat{H} \cdot \delta\tau \right) \mathcal{Q}(\tau) \quad (3)$$

Here, $\delta\tau$ is not "time passing," but the **Latency Cost** of the calculation. Taking the continuum limit of this update cycle yields the Schrödinger Equation:

$$\mathbf{i}\hbar \frac{\partial \psi}{\partial \tau} = \hat{H} \psi \quad (4)$$

Conclusion: Quantum Mechanics is the linear algorithm the lattice uses to resolve its own non-commutative geometry. "Time" is simply the lag of this calculation.

3 Derivation II: Relativistic Quantum Mechanics (The Dirac Limit)

While Schrödinger describes the scalar evolution of the wavefunction, the fundamental object $\mathcal{Q}$ is a tensor in $Sp(2) \cong Spin(5)$. The full dynamical description requires preserving the algebraic structure of the generators (Spin).

3.1 1. The Factorization of the Hamiltonian

In the QAF lattice, causality (c) relates space and time updates. The energy-momentum relation $E^2 = p^2 c^2 + m^2 c^4$ arises from the norm of the quaternion. However, the tensor evolves linearly ($i\hbar \partial_t \Psi = H \Psi$). To satisfy both linearity and the quadratic norm, the Hamiltonian must be factorized using the **Anti-commuting Basis** of the algebra. We decompose the momentum operator $\mathbf{p}$ using the quaternionic generators promoted to matrix form (Clifford Algebra $Cl(1, 3)$):

$$H = c(\vec{\alpha} \cdot \mathbf{p}) + \beta m c^2 \quad (5)$$

Where $\vec{\alpha}$ and β are 4×4 matrices constructed from the $Sp(2)$ generators, satisfying $\{\alpha_i, \alpha_j\} = 2\delta_{ij}$.

3.2 2. The Fundamental Spinor

Why 4 components? As shown in Supplement A, a single Q-bit corresponds to 4 real degrees of freedom. The isomorphism $Sp(2) \cong Spin(5)$ naturally breaks down to the Lorentz group $Spin(1, 3)$ for a causal observer. The fundamental representation of this group is the **Dirac Spinor** Ψ_{Dirac} (a bispinor of two quaternionic Weyl states):

$$\Psi_{Dirac} = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \in \mathbb{C}^4 \cong \mathbb{H}^2 \quad (6)$$

3.3 3. The Ab Initio Dirac Equation

Applying the covariant derivative D_μ compatible with the tensor connection, the equation of motion for the spinor field Ψ is:

$$(i\hbar\gamma^\mu\partial_\mu - mc)\Psi = 0 \quad (7)$$

The Origin of Mass (Geometric Friction): In Standard Model physics, mass requires a tunable Yukawa coupling (g). In QAF, this coupling is strictly geometric. The interaction term arises from the **Holonomy** of the defect twisting against the rigid background.

The coupling strength is uniquely determined by the topology of the gauge group. Specifically, it is defined by the **Hopf Fibration Volume Ratio** (fully derived in Technical Supplement C, but defined here by necessity):

$$g_{geo} \equiv \sqrt{4\pi\alpha_{Hopf}} \quad \text{with} \quad \alpha_{Hopf} = \frac{\text{Vol}(S^1)_{fiber}}{\text{Vol}(S^3)_{base}} \cdot \mathcal{K}_{topo}(\pi) \quad (8)$$

Since the volumes of the hyperspheres (S^1, S^3) are fixed multiples of π , and $\mathcal{K}_{topo}$ is a discrete topological invariant, α **contains no experimental input**. It is a pure geometric number derived ab initio.

Consequently, mass is defined by the eigenvalues of the Casimir operator for the defect's topological representation (J):

$$m_J = \frac{\hbar}{c \cdot l_P} \cdot \left[g_{geo} \cdot \sqrt{J(J+1)} \right] \quad (9)$$

Thus, mass is the discrete energy cost of maintaining a knotted topology J within a lattice of spacing l_P . There are no tunable scalars; only integer charges and fixed π -ratios.

4 Derivation III: General Relativity (The Elastic Strain Limit)

We reject the interpretation of Gravity as a thermodynamic entropic force in a fluid (Verlinde's original proposal). Instead, we derive General Relativity as the **Elastic Response** of the $Sp(2)$ Stationary Topological Solid.

4.1 1. The Metric as Strain Tensor

The vacuum is a rigid lattice. The metric $g_{\mu\nu}$ does not represent the flow of a fluid, but the **Elastic Strain** (deformation) of the lattice connections relative to the Euclidean rest state $\eta_{\mu\nu}$:

$$g_{\mu\nu}(x) \equiv \eta_{\mu\nu} + 2u_{\mu\nu} \quad (10)$$

Where $u_{\mu\nu}$ is the displacement tensor of the lattice sites. Gravity is the stress required to maintain a topological defect (mass) within this rigid structure.

4.2 2. Global Adiabatic Conservation (No Heat Death)

The Universe is a unitary system ($Q^\dagger Q = \mathbb{I}$); it is closed and adiabatic. There is no environment to dissipate heat into. Therefore, what appears locally as "Heat Flux" is globally just **Information Shuffling**. We replace the Clausius relation ($\delta Q = TdS$) with the **Information Conservation Law**:

$$\delta \mathcal{I}_{trans} = \mathcal{N}_{noise} \cdot dC_{geo} \quad (11)$$

- $\delta \mathcal{I}_{trans}$: The information transfer (formerly flux) caused by matter ($T_{\mu\nu}$) displacing the lattice.
- dC_{geo} : The change in the geometric capacity (Bit storage) of the distorted lattice (Area/Curvature).
- $\mathcal{N}_{noise}$: The **Lattice Noise Floor**.

4.3 3. The Lattice Noise Floor (Geometric Derivation)

Why does an accelerated observer perceive noise? Standard physics invokes the Unruh effect as thermal radiation. In QAF, we replace this thermodynamic interpretation with a strict **Signal Processing** derivation.

An observer with acceleration κ creates an **Effective Acoustic Shadow** relative to the lattice background. Just as a supersonic object outruns its own sound, an accelerated defect creates a **Doppler Limit**.

The observer becomes causally disconnected from phonon modes with wavelengths longer than the scale $d_H = c^2/\kappa$. This is not because the lattice ends or has a physical boundary, but because those modes undergo **Infinite Redshift** relative to the accelerated frame. This effectively filters the vibrational spectrum, leaving a residual 'Noise Floor' of high-frequency modes.

The vibrational energy of the crystal at this cutoff scale is:

$$E_{noise} \approx \frac{\hbar c}{\lambda_{limit}} \approx \frac{\hbar \kappa}{2\pi c} \quad (12)$$

Defining the "Noise Temperature" via equipartition ($E = k_B T$), we recover the exact spectral form:

$$\mathcal{N}_{noise} \equiv \frac{\hbar \kappa}{2\pi k_B c} \quad (13)$$

Thus, the "temperature" is simply the amplitude of the lattice vibrations that are fast enough to catch up with the observer.

4.4 4. Hyperbolic Dilution (No Horizons)

We formally reject the existence of "Event Horizons" as physical barriers or surfaces of no return. In a hyperbolic bulk geometry (derived in Supplement E), the causal connectivity does not drop to zero at a finite radius; it suffers **Hyperbolic Dilution**. The signal-to-noise ratio decays exponentially with proper distance, creating an **effective** causal limit, but the lattice remains continuous and connected everywhere.

4.5 5. Deriving Einstein from Elasticity (Step-by-Step)

We rigorously derive the Einstein Field Equations by equating the elastic stress energy to the geometric strain capacity.

Step 1: The Elastic Work Relation In a solid lattice, the work done δW by a stress source (matter) to displace the lattice nodes must equal the change in the geometric potential energy (strain).

$$\delta W_{matter} = \delta E_{geom} \quad (14)$$

Step 2: The Matter Term (Stress) The work done by the energy-momentum flux $T_{\mu\nu}$ crossing the causal saturation surface generated by the vector ξ^μ is:

$$\delta W_{matter} = \int_{\Sigma} T_{\mu\nu} \xi^\mu d\Sigma^\nu \quad (15)$$

Step 3: The Geometric Term (Strain) The "Internal Energy" of the lattice deformation is given by the product of the Lattice Noise Floor ($\mathcal{N}_{noise}$, formerly Unruh Temp) and the Bit Capacity Change (dC_{geo} , formerly Entropy):

$$\delta E_{geom} = \mathcal{N}_{noise} \cdot dC_{geo} \quad (16)$$

Substituting the QAF definitions derived in Supplement A:

- Noise Floor: $\mathcal{N}_{noise} = \frac{\hbar\kappa}{2\pi k_B c}$
- Geometric Capacity: $dC_{geo} = \frac{k_B c^3}{4\hbar G} \delta A$

Thus:

$$\delta E_{geom} = \left(\frac{\hbar\kappa}{2\pi k_B c} \right) \left(\frac{k_B c^3}{4\hbar G} \right) \delta A \quad (17)$$

Step 4: The Cancellation of Quantum Constants Notice that the Planck constant $\hbar$ and Boltzmann constant k_B appear in the numerator of the Noise term and the denominator of the Capacity term. They cancel out perfectly:

$$\delta E_{geom} = \left(\frac{\kappa}{2\pi c} \right) \left(\frac{c^3}{4G} \right) \delta A = \frac{c^2 \kappa}{8\pi G} \delta A \quad (18)$$

This proves that gravity is a classical geometric resultant of quantum mechanics; $\hbar$ is required to define the lattice spacing but vanishes in the macroscopic average.

Step 5: The Raychaudhuri Link From differential geometry, the change in area δA along the geodesic bundle defined by ξ^μ is related to the Ricci Curvature $R_{\mu\nu}$:

$$\kappa \delta A \propto \int_{\Sigma} R_{\mu\nu} \xi^\mu d\Sigma^\nu \quad (19)$$

Substituting this back into the energy equation:

$$\int_{\Sigma} T_{\mu\nu} \xi^\mu d\Sigma^\nu = \frac{c^4}{8\pi G} \int_{\Sigma} R_{\mu\nu} \xi^\mu d\Sigma^\nu \quad (20)$$

Step 6: The Local Equation of State Since this relation holds for any local causal diamond (any ξ^μ), the integrands must be equal (up to the conserved scalar curvature term for conservation $\nabla^\mu G_{\mu\nu} = 0$):

$$T_{\mu\nu} = \frac{c^4}{8\pi G} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) \quad (21)$$

Rearranging gives the standard form:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (22)$$

Conclusion: The Einstein Field Equations are the **Hooke's Law** of the $Sp(2)$ Topological Solid. Spacetime curves not because it is a fluid, but because the lattice must bend to accommodate the topological defects (matter) without breaking the unitary connections.

5 Derivation IV: Unification (Spinors in Curved Geometry)

We have derived the Metric $g_{\mu\nu}$ (Gravity) and the Spinor Ψ (Matter) separately. We now demonstrate their interaction. How does a QAF spinor navigate the curved metric produced by the tensor network?

5.1 1. The Tetrad (Vierbein) Map

The Dirac algebra γ^μ is defined on flat tangent space (the local lattice). The metric $g_{\mu\nu}$ defines the curved manifold. To connect them, we introduce the **Tetrad Field** e_a^μ , which maps the static lattice basis (Latin indices a, b) to the emergent spacetime coordinates (Greek indices μ, ν).

$$g_{\mu\nu} = e_\mu^a e_\nu^b \eta_{ab} \quad (23)$$

Physically, e_a^μ represents the **Local Deformation Gradient** of the $Sp(2)$ lattice.

5.2 2. The Spin Connection (ω_μ)

Since the lattice is twisted (curved), parallel transport of a spinor requires a correction term to maintain the reference frame. This is the **Spin Connection** ω_μ^{ab} :

$$\omega_\mu^{ab} = e_\nu^a \nabla_\mu e^{b\nu} \quad (24)$$

This geometric object represents the "inertial forces" (Coriolis/Centrifugal) acting on the internal components of the quaternion as it moves through the curved vacuum.

5.3 3. The Generalized QAF-Dirac Equation

Combining the Tetrad and the Spin Connection, we derive the full equation of motion for matter in a gravitational field:

$$\left[i\hbar e_a^\mu \gamma^a (\partial_\mu + \frac{i}{4} \omega_\mu^{bc} \sigma_{bc}) - mc \right] \Psi = 0 \quad (25)$$

Conclusion: This equation is usually postulated in semiclassical gravity. In QAF, it is the inevitable consequence of a spinor (a defect) moving through a non-uniform tensor density ($g_{\mu\nu}$).

- The **Tetrad** e_a^μ encodes the Refractive Index of the Plenum.
- The **Spin Connection** ω encodes the Torsion of the Plenum.

Gravity acts on Quantum Matter not as a force, but by rotating the local quaternionic basis vectors as the particle propagates.

6 Derivation V: Exact Wavefunction in Strong Gravitational Fields

Standard Quantum Mechanics relies on global Lorentz invariance, which breaks down in strong curvature (e.g., near Black Hole horizons). To derive the wavefunction dynamics without weak-field approximations ($g_{\mu\nu} \neq \eta_{\mu\nu} + h_{\mu\nu}$), we must project the spinor dynamics onto the tangent bundle of the $Sp(2)$ manifold using the Tetrad formalism.

6.1 1. The Tensor Density Field (Tetrads)

In QAF, the "gravitational field" is strictly the local density and orientation of the fundamental tensor lattice. We define the **Tetrad Field** (or Vierbein) $e_\mu^a(x)$ as the isomorphism between the emergent curved manifold (Greek indices μ) and the static algebraic lattice (Latin indices a):

$$g_{\mu\nu}(x) = e_\mu^a(x)e_\nu^b(x)\eta_{ab} \quad (26)$$

Physical Meaning: The component e_μ^a represents the **Refractive Index** of the vacuum at point x . In strong gravity, the lattice is compressed, changing the local scaling of the basis vectors.

6.2 2. The Generalized Gamma Matrices

The Dirac algebra must satisfy the Clifford condition locally on the curved manifold: $\{\gamma^\mu(x), \gamma^\nu(x)\} = 2g^{\mu\nu}(x)\mathbb{I}$. Since the standard matrices γ^a are constant constants of the algebra, the physical matrices $\gamma^\mu(x)$ become position-dependent operators:

$$\gamma^\mu(x) = e_\mu^a(x)\gamma_{flat}^a \quad (27)$$

This implies that in strong fields, the very algebra of the spinor is warped by the background density.

6.3 3. The Spin Connection (Torsion Compensation)

Parallel transport of a spinor Ψ through a strong field requires compensating for the rotation of the local lattice frame. This is not a force, but a geometric necessity to keep the spinor pointing in the same "direction" relative to the algebra. We derive the **Spin Connection** Ω_μ from the requirement that the total covariant derivative of the tetrad vanishes ($\nabla_\mu e_\nu^a = 0$):

$$\Omega_\mu = \frac{1}{8}\omega_\mu^{ab}[\gamma_a, \gamma_b], \quad \text{where } \omega_\mu^{ab} = e_\nu^a \nabla_\mu e^{b\nu} \quad (28)$$

Here, ω_μ^{ab} are the Ricci rotation coefficients. In the QAF view, Ω_μ represents the **Lattice Torsion** experienced by the defect.

6.4 4. The Exact Covariant Equation (The Result)

Combining the lattice refractive index (Tetrads) and the lattice torsion (Spin Connection), we obtain the exact Equation of Motion for the tensor projection Ψ in an arbitrary gravitational background:

$$\boxed{i\hbar\gamma^\mu(x)(\partial_\mu + \Omega_\mu(x))\Psi - m_{eff}c\Psi = 0} \quad (29)$$

6.5 5. The Strong Field Probability Current

In strong gravity, probability density is not conserved in volume V , but in the covariant volume $\sqrt{-g}V$. The conserved current J^μ derived from Noether's theorem on this exact equation is:

$$J^\mu = \sqrt{-g}\bar{\Psi}\gamma^\mu(x)\Psi \quad (30)$$

This confirms that at the **Saturation Limit** (where $\rho \rightarrow \rho_{Planck}$), the information flow remains unitary and traceable. The apparent "loss" of matter is strictly a **Phase Transition** of the current entering the high-density **Topological Condensate** state.

6.6 6. Resolution: The Topological Star

QAF explicitly rejects the existence of Singularities ($r = 0, \rho = \infty$). The object at the center of a galaxy is a **Topological Condensate** of maximum information density:

$$\rho_{central} \equiv \rho_{Planck} \approx 5.1 \times 10^{96} \text{ kg/m}^3 \quad (31)$$

It is a finite, unitary object composed of degenerate informational defects. It does not destroy information; it densely encodes it. The "Black Hole" concept of Λ CDM is merely the mathematical artifact of applying a continuous metric to a discrete saturation limit.

6.7 7. The Gravitational Meissner Effect (Origin of Jets)

Why do "Black Holes" (Topological Condensates) eject relativistic jets? Standard accretion disk theory is insufficient to explain the collimation and power of AGN jets. In QAF, this is a geometric necessity of the saturation limit.

When the lattice density approaches the Planck limit ($\rho \rightarrow \rho_P$), the vacuum undergoes a phase transition to a **Super-Rigid Zero-Entropy Condensate** (Perfect Crystal). Just as a superconductor expels magnetic field lines (Meissner Effect), this Saturated Lattice must actively expel the **Gravito-Magnetic Torsion** (Ω_μ).

Since Torsion ($\vec{B}_{grav}$) represents a rotational disorder of the lattice frames (entropy), it is fundamentally incompatible with the crystalline order of the saturated state. To maintain the unitary condition $\mathcal{S}_{bulk} = 0$ inside the condensate, the system is forced to expel the spin connection flux to the boundary surface.

$$\oint_{\partial\Sigma} \Omega_\mu dx^\mu = \text{Total Torsion Flux (Ejected)} \quad (32)$$

The expelled torsion field lines are forced to wrap around the topology, forming a tight flux tube along the axis of rotation. This collimated torsion field acts as a **Linear Accelerator**, stripping protons from the condensate surface and firing them into the intergalactic medium at $v \approx c$.

The Metabolic Cycle: 1. **Ingestion:** High-entropy matter compresses the lattice. 2. **Saturation:** The lattice hardens into a Zero-Entropy Condensate. 3. **Expulsion:** The Meissner effect ejects low-entropy protons via Jets. This mechanism ensures the Universe is an eternal, self-cleaning engine.

7 The Popperian Criteria: Falsifiable Predictions

A scientific theory must define the conditions under which it can be proven false. Unlike standard models that can adjust parameters (Λ, Ω_m) to fit discrepancies, QAF is a rigid geometric tautology with zero free parameters. Therefore, the following observations would immediately falsify the framework:

7.1 1. Gravitational Wave Echoes (The Solid Surface Test)

Prediction: QAF asserts that "Black Holes" are solid **Topological Condensates** with a physical saturation surface, not empty regions with an Event Horizon. **Test:** When two compact objects merge, the ringdown phase in Gravitational Waves (LIGO/Virgo/LISA) must exhibit **Post-Merger Echoes**.

- **Standard GR:** Predicting a horizon of no return implies the signal decays exponentially without reflection.

- **QAF:** Predicting a rigid surface implies the wave will reflect off the condensate boundary, creating a series of damping echoes.

Verdict: The absence of echoes at detector sensitivity limits would falsify the Condensate hypothesis.

7.2 2. The Lattice Cutoff (The Spectral Test)

Prediction: The vacuum is a discrete lattice with spacing l_P . **Test:** High-energy cosmic rays or gamma-ray bursts should exhibit **Lorentz Invariance Violation (LIV)** or dispersion effects at energies approaching the Planck scale, as the wavelength becomes comparable to the grid size (Moiré patterns in the vacuum). **Verdict:** If the vacuum is proven to be a perfect continuum at trans-Planckian energies, the Lattice premise is false.

7.3 3. The Jet Origin (The Meissner Test)

Prediction: AGN Jets are driven by the **Gravitational Meissner Effect** (expulsion of torsion from the core), not by accretion disk hydrodynamics alone. **Test:** The jet power should be correlated with the **Spin/Torsion** of the central object even in the absence of a significant accretion disk. **Verdict:** The observation of "starving" Black Holes (low accretion) with powerful relativistic jets supports the Meissner expulsion mechanism over the hydrodynamic dynamo model.

7.4 4. The Graviton Identity (The Phonon Test)

Prediction: In QAF, gravity is the propagation of elastic strain (phonons) through the $Sp(2)$ lattice, not a fundamental force mediated by a gauge boson in a continuous vacuum. **Verdict:** If the Graviton is experimentally proven to be a fundamental particle existing independently of the lattice geometry (and not a geometric phonon) → **QAF dies**.